

Accountable Anonymity for Public Money (DRAFT v2)

Identity-Bound Bearer Payments with No Trusted Opener

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Public money is going digital, and every design on the table pays for the convenience with a ledger its issuer can read, freeze, and mine. Cash resists surveillance, but only by keeping no evidence at all: no proof of payment, no recourse, no trail. This paper presents BUCK Notes, a bearer-payment layer for an identity-aware stablecoin that refuses both defaults. No one can trace payments in bulk—not an analytics firm, not an omnibus subpoena, not even the system’s own KYC issuer. Yet the two parties to any payment can always prove exactly who the other was, with evidence sound enough to convict on. The guarantee is the **non-deniable receipt**: even a *colluding* payer and payee cannot complete a payment without each holding evidence that names the other.

The consequences read like due process implemented in mathematics (the **compulsion calculus**): evidence of every payment exists; only the participants hold it; each may surrender, refuse, or destroy their copy; and no institution can be compelled to produce what it provably never had. One design choice makes this work: payments bind KYC-certified *identities*—not just account keys—(**Identity-M binding**¹). A single re-encryption gadget, instantiated three ways, yields every realisable note flavour (addressed or bearer, public or private issuer). All three are shipped and measured on Ethereum. We prove the fourth flavour impossible, show that the obvious cheap binding is vacuous, and prove that zero-knowledge registry membership is necessary for the doubly-anonymous case under a stated restriction. This is exactly the branch the shipped system takes. Every theorem carries an executable witness. (PDF, Text)

¹The binding of payments to a KYC-certified identity point M , rather than to disposable account keys.

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DRAFT

1 Introduction

The choice a society makes about the privacy of its money is a choice about the balance of power between the individual and the institutions that watch over them. This paper is about a way to make that choice deliberately: payments no institution can read in bulk, where the two parties (and only they) always hold court-grade proof of who dealt with whom.

That choice is currently being made by default. Every central-bank digital currency in design or deployment couples digital convenience to a ledger some institution reads in bulk; the visibility is not an engineering accident; it is the pitch to the governments funding it². Cash is the only widely used instrument that resists bulk observation, and it resists by carrying no information at all: no issuance trail, no proof a court can rely on, no recourse when stolen. The public debate treats these as the only two options: total legibility or total opacity. Forty years of cryptography has not yet dislodged that false dichotomy from policy.

Chaum asked the right question in 1982³. His answer stalled on institutional grounds, not cryptographic ones. The anonymity line that followed (Zerocash and its descendants, mixers, and privacy pools^{4, 5, 6}) made payments untraceable and unaccountable in one stroke. The auditable-CBDC line (PRCash, UTT, PEReDi, Platypus^{7, 8, 9, 10}) restored accountability by giving a designated authority (auditor, committee, or central bank) the power to open transactions. No prior system provides both untraceable payments and mandatory accountability with *no opener at all*.

This paper presents a system that does, and locates precisely what it costs. The design goal is civic before it is cryptographic:

- **Justice must work.** When a payment is implicated in a crime, the legal system must be able to obtain cryptographically sound evidence of exactly who paid whom (strong enough to convict on, impossible to forge against the innocent). And the evidence must be guaranteed to *exist*: two colluding criminals must not be able to arrange an untraceable payment.
- **Compulsion must be targeted.** That evidence is held by (and can be demanded from) only the people party to the payment. There is no registry, exchange, issuer, or validator that a warrant, a breach, or a purchase can turn into bulk surveillance, because no such party holds the information in any usable form.
- **Refusal must be possible.** If the legal order itself turns unjust (an authoritarian government criminalising beliefs, associations, donations), a citizen retains the physical ability to refuse: to withhold, or irreversibly destroy, their own keys. The data then stays private against any coalition of compelled institutions and archived chain history. Surveillance of the unwilling is not available at any institutional price; it requires coercing human beings, one at a time, in the open.

²Bank for International Settlements. "Rise of the Central Bank Digital Currencies." [TODO]

³Chaum, D. "Blind Signatures for Untraceable Payments." CRYPTO 1982. [TODO exact cite]

⁴Ben-Sasson, E., et al. "Zerocash: Decentralized Anonymous Payments from Bitcoin." IEEE S&P 2014. [TODO]

⁵Pertsev, A., Semenov, R., Storm, R. "Tornado Cash." 2019. [TODO]

⁶Buterin, V., Schaefer, J., Tromer, E., et al. "Blockchain Privacy and Regulatory Compliance: Towards a Practical Equilibrium (Privacy Pools)." 2023. [TODO]

⁷Wuest, K., Kostianen, K., Capkun, V., Capkun, S. "PRCash: Fast, Private and Regulated Transactions for Digital Currencies." FC 2019. [TODO verify authors/title]

⁸Tomescu, A., et al. "UTT: Decentralized Ecash with Accountable Privacy." IEEE S&P 2022. [TODO verify]

⁹Kiayias, A., Kohlweiss, M., Sarencheh, A. "PEReDi: Privacy-Enhanced, Regulated and Distributed Central Bank Digital Currencies." ACM CCS 2022. [TODO verify]

¹⁰Wuest, K., Kostianen, K., Delius, N., Capkun, S. "Platypus: A Central Bank Digital Currency with Unlinkable Transactions and Privacy-Preserving Regulation." ACM CCS 2022. [TODO verify]

These three goals are usually presumed to be in tension. The central claim of this paper is that they are simultaneously satisfiable, practically, on a public blockchain today.

What makes all three possible is one *counter-intuitive inversion*: payments must bind *identities*—certified once, at KYC, and held by persons—not just the *account keys* that hold funds. We call this **Identity-M binding**. Everything else follows from it: the receipt that always exists, key recovery without any custodian, the impossibility of one whole quadrant of the design space, and the exact place where accountability stops being cheap.

The unexpected result of adding *more* identity binding to the blockchain is that *less* of our identity leaks, and *more* of our rights and privacies are protected.

1.1 Contributions

Seven contributions, in the order the paper builds them:

- **The non-deniable-receipt invariant** (Def. 2): a security definition for payment systems whose new clause is collusion resistance: even a *willing* payer and payee cannot construct an accepted payment with no receipt. Accountability is **bilateral**, held by the counterparties; there is no system-wide opener.
- **The compulsion calculus** (Sec. 6.4): the invariant, composed with issuer-blindness (the KYC issuer cannot trace even its own certified clients; Thm. 3), yields three properties we believe are jointly new for a payment system: evidence *exists* for every payment; *only* the direct participants hold it; and it is *refusable*: without a participant’s keys, surrendered or compelled, the ledger is computationally silent, under DDH, for as long as that assumption stands. We argue this is the right cryptographic substrate for due process, and the right failure mode under unjust process.
- **Identity-M binding** (Sec. 7): the inversion that makes the rest practical. One gadget (prove an encryption of a registered Identity under a target key) instantiated three ways covers the whole realisable design space. Note spend authorization keys on identity, so a lost account key is recoverable with no custodian, and the binding survives the time gap between an anonymous payer and an anonymous payee.
- **A complete, shipped construction** of all three realisable flavours – addressed/public (A1), addressed/private (A2), bearer/public (B1) – on Ethereum: Groth16 batch mint, shared flavour-bound spend, and spend-time composition of a deposit sigma, a bound set-membership proof, and a note-to-ciphertext tie, every verifier real and measured (Sec. 12).
- **The forced taxonomy** (Thm. 8): the fourth flavour—a bearer note from a hidden issuer—is impossible, by an information-theoretic argument.
- **The cost of anonymity, located** (Sec. 11): the receipt is nearly free when one party is named at the binding moment. When both are anonymous, the obvious cheap coupling is *vacuous* (Lem. 16), and a collusion-resistant receipt requires zero-knowledge registry set-membership under a single-binding-step restriction (Prop. 17). The shipped system takes exactly the branch the proposition permits, and we measure it.
- **A reproducible artifact**: contracts, circuits, and verifiers, open and test-covered—over 400 Foundry tests, 330 wallet tests, an end-to-end suite where every verifier is the real Groth16 artifact —and the paper’s own claims carry executable witnesses (Appendix 14) that re-run on each build.

2 The Shape of the System, in Plain Language

Before the formalism, the experience.

Alice owes Bob one hundred BUCKs. She opens her wallet and mints a *note*: her wallet burns one hundred of her BUCKs into a shared pool and publishes a single opaque number (a cryptographic commitment) onto the public chain. The commitment reveals nothing: not the amount, not Bob, not (for the private flavour) even that it was Alice. Alice then hands Bob the note's *opening* (a few hundred bytes) by any channel she likes: a message, a QR code on her phone's screen, a slip of paper. Days or months later, from any account he controls, Bob *deposits* the opening: his wallet proves (in *zero knowledge*¹¹), revealing nothing about which note, that some unspent note in the pool is his to claim, and the pool pays him one hundred BUCKs. An observer watching the chain sees a commitment appear at one moment and an unrelated-looking payout occur at another, with nothing connecting them – not the amounts, not the parties, not the timing.

Three further facts make this different from a mixer (a pool that breaks payment trails and does nothing else), and they are the subject of this paper.

First, the payment writes its own sealed receipt. Built into every note (enforced by the chain's verifiers, impossible to omit) is the cryptographic material from which each party can produce a proof naming the *other party's certified legal identity*. Bob can prove "this payment came from Alice"; Alice can prove "Bob is the one who deposited it." The proof is sound: nobody can manufacture a receipt naming someone who was not actually party to the payment. And it is *unavoidable*: Theorem 13 shows that even if Alice and Bob conspire – both willing, both technically sophisticated – they cannot construct an accepted payment whose receipt material is missing or unusable. The worst a conspiracy can do is render a note addressed to nobody, and such a note is *unspendable*: the pool will not pay out on it.

Second, only Alice and Bob hold the seal. The receipt is not deposited with any registrar. It does not exist in any database. The KYC office that once certified Alice's identity cannot reconstruct her payments; we prove (Theorem 3) that even a *fully breached* identity office, holding every certificate it ever issued plus the entire blockchain, gains only a negligible advantage, under DDH, in learning who transacted with whom. A court that needs the receipt has exactly one place to get it: from Alice, or from Bob, by lawful order directed at that person.

Third, the seal answers to its holders—and only its holders. Served with a lawful order, Bob can comply by surrendering his receipt. The court then obtains evidence (strong enough to convict on) that mathematics—not testimony—ties to Alice. This serves justice: the victim of a fraud or the honest counterparty of a criminal can always prove exactly whom they dealt with.

Yet if the order comes from an unjust regime (e.g., criminalising a donation, membership, or purchase), Bob can refuse. He can decline to produce his key (accepting the personal consequence) or destroy it beforehand as standard practice. Then no one on earth can open that payment—not the identity office, not the chain operators, not even a full server seizure. The mathematics simply does not know who is in power.

This is, we will argue, the right shape for the money of a free society: per-payment accountability *stronger* than paper cash (which offers none), surveillance resistance *stronger* than bank money (which offers none), and a default the citizen controls with their own hands. Cash with a carbon-copy receipt, where the only copies belong to the people who made the payment.

The system comes in three flavours, fixed by who is named when:

- **A1: addressed cheque, public payer.** The note names its recipient cryptographically

¹¹A cryptographic technique allowing one party to prove a statement is true without revealing any information beyond the validity of the statement itself.

(only Bob’s identity can deposit it; a thief who photographs the paper gains nothing) and its issuer openly (payroll, invoices, audited disbursements).

- **A2: addressed cheque, private payer.** As A1, but Alice’s identity is hidden from everyone except Bob; the flavour for paying privately while still handing the recipient a receipt that names you. This is the cryptographically expensive flavour; Sections 11–12 locate the cost and explain it.
- **B1: bearer cheque, public payer.** Whoever holds the opening can deposit it (walking-around money; the paper surface), and the depositor’s identity is delivered (encrypted, bilaterally) to the issuer. The fourth combination, a bearer note from a hidden issuer, is not merely undesirable; it is impossible (Theorem 8).

The design space is therefore the four combinations of *addressed vs. bearer* and *public vs. private issuer*:

	Public issuer (named in clear at mint)	Private issuer (hidden at mint)
Addressed (recipient named at spend via coupling)	A1: cheap (batch Schnorr)	A2: requires ZK set-membership
Bearer (depositor named at spend)	B1: cheap (depositor binding)	B2: impossible

When both parties must stay anonymous at the moment their identity is bound (the lower-right cell), cheap couplings are vacuous (Sec. 11). Set-membership in the registry accumulator becomes necessary. This is the source of the "cost of anonymity" measured later. (The upper-left and lower-left cells are the cheap cases; see 3 for why the binding moments matter.)

3 Binding Moments and the Split Handshake

A direct transfer between two registered accounts performs the bilateral identity handshake in a single on-chain call: both parties' identity anchors are live at the same moment. A note payment deliberately splits that handshake across time. Alice mints (naming or hiding the payer) now; Bob deposits (naming or hiding the payee) later. Unlinkability forbids any public value from bridging the two moments. This *split-handshake* or *binding-moment asymmetry* is the root technical problem the rest of the design must solve.

4 The Identity Fountain

A single KYC ceremony produces one scarce root of trust per person: an identity scalar $m = H(\text{id})$ and a Pointcheval-Sanders¹² signature on it. From that master credential the owner can, entirely offline, derive an arbitrary number of fresh on-chain accounts:

- A new key pair (sk, pk) .
- A fresh ElGamal¹³ encryption $E = (rG, M + r \cdot pk)$ of the same $M = mG$.
- A blinded *presentation* $(A, B) = (a\sigma_1, a\sigma_2 + byG)$ of the PS signature, with a proof that it presents a credential on the same m the ciphertext encrypts.

The presentation is verifiable as a credential from the same issuer, yet it is a uniform pair of points whatever the identity: it carries no information linking it to any other derived account or to the original KYC record, even to a party holding the identity scalar. (A rerandomised signature alone would not do: it remains a valid signature on m , which the issuer, or any past counterparty, verifies directly. The presentation is the A' design of the *Proofs* document, Theorems 1'-2'; an executable two-sided witness – the old pair identified under the true m , the presentation under none – is in the *Examples* document.) Even the issuer itself, holding every signature it ever issued plus the entire blockchain, cannot cluster the on-chain accounts back to the same person: the credential side is hidden unconditionally, the ciphertext side under DDH (Theorem 3). The only people who can recover M are those to whom it is deliberately disclosed (via the bilateral approve handshake or a note payload); they also receive the off-chain plaintext id so they can confirm it matches the issuer's certification.

This "one KYC, unlimited unlinkable personas" capability (the *Identity Fountain*) is what supplies stable identity anchors across the mint-to-spend time gap without forcing all of a person's activity to be linkable. It is the mechanism that makes issuer-blindness (Theorem 3) and the targeted nature of compulsion practical. The presentation's security argument is stated in the *Proofs* document and awaits independent review; the results below that rest on it are conditional on that review.

¹²Pointcheval-Sanders (PS) signatures: a type of structure-preserving signature that supports efficient rerandomization, allowing the holder to produce fresh versions of the credential without the issuer's involvement. On their own such versions remain recognisable to anyone holding the signed message, so Alberta Buck publishes a blinded presentation with a proof (the A' design, *Proofs* Part I) rather than a rerandomised signature.

¹³ElGamal encryption: a public-key encryption scheme over elliptic curve points that supports re-randomization and is semantically secure under the DDH assumption.

5 Related Work

Every predecessor gives up one of the two goods. The anonymity line gives up accountability; the auditable line gives up doing without an opener. No prior system provides a *mandatory, bilateral, collusion-resistant* receipt with *no system-wide opener*:

System	Unlinkable	Mandatory receipt	Collusion-resistant	Trusted opener?	Bearer/paper
Chaum e-cash ³	issuer-blind	no	no	central bank	no
Zerocash / Zcash ⁴	yes	no (opt. viewkey)	no	no	no
Tornado / Privacy Pools ⁶	yes	no (innocence)	no	no	no
PRCash ⁷	yes	regulator-facing	via auditor	yes (auditor)	no
UTT ⁸	yes	budget-gated	via committee	yes (committee)	no
PEReDi ⁹	yes	regulator-facing	via threshold	yes (maintainers)	no
Platypus ¹⁰	yes	regulator-facing	via central bank	yes (cen. bank)	no
CBDC (e-CNY, digital euro) ²	no	n/a	n/a	issuer sees all	no
BUCK Notes	yes	yes	yes	no	yes

The auditable-privacy line (PRCash, UTT, PEReDi, Platypus) is the closest comparison. All four restore accountability to unlinkable payments, and all four do it by appointing an authority (an auditor, a committee, the central bank) with the power to de-anonymise. That authority is exactly what a breached, purchased, or repurposed institution turns into a population-scale surveillance instrument. BUCK’s accountability is *counterparty-held*. The only institutional anchor is the per-identity KYC issuer, which (Theorem 3) can map a *surrendered* identity to a person but can never produce the link itself: compulsion runs depth-first, through named individuals – never breadth-first, across the system.

On the anonymity line, Zerocash-style view keys are *optional* and payer-controlled – precisely not mandatory – and the innocence-proving of Privacy Pools is voluntary exculpation, not compellable evidence. On the bearer axis, prior physical-form digital cash (Casascius coins, BIP-38 paper wallets) wraps a bearer secret with no identity semantics at all. BUCK’s addressed flavours make the paper artifact itself useless to anyone but the named recipient.

6 Model, Adversaries, and the Invariant

The terms, in brief: a person’s *Identity* is a single curve point derived from their KYC record; *accounts* are disposable key pairs, each carrying an encrypted copy of its owner’s Identity and a credential that proves certification without identifying the certificate; and a *registry* accumulates the set of certified Identities so that membership in it can later be proven in zero knowledge.

Formally (see Sec. 4 for the mechanism): an Identity is a KYC-certified point $M = mG \in \mathbb{G}_1$ (BN254), where the scalar $m = H(\text{id})$ is a preimage-resistant hash of the certified attribute record id . Using the Identity Fountain, a person derives and registers any number of on-chain accounts. Each account X stores its owner’s Identity ElGamal-encrypted under its own key as $E_{\text{addr}}[X] = (R_X, C_X) = (r_X G, M_X + r_X pk_X)$ with $pk_X = sk_X G$. Its registration transcript carries a blinded presentation (A, B) of the client’s Pointcheval-Sanders (PS) credential over m , with a proof of validity: a pair that no holder of m , the issuer included, can test (the rerandomised signature an earlier design published was recognisable to anyone holding m). Membership is a two-level Poseidon accumulator: each registry certifies salted leaves into its own subtrees, an aggregator composes the subtree roots, and its root rt_{id} is posted into a ring whose age each consumer bounds. Membership is provable in zero knowledge, and a party holding every certified identity scalar decides none of it, because the leaves are salted from wallet secrets. Admission is the registry’s certification: **register** refuses a caller-supplied leaf.

These two secrets carry different authority. The identity scalar m grants the capability to *name* (verify receipts, name a counterparty) and is disclosed to every counterparty by design – so it decrypts nothing: notes are addressed to a separate receiving key $pk_{\text{recv}} = kG$, derived from wallet seed material and certified beside the identity in one accumulator leaf: Bob learns Alice’s m when she pays him, and vice versa. The PS credential σ , in contrast, grants *write* capability (register accounts, authorize spends) and is held only by its owner. The system’s security rests on the fact that m alone, without σ , is insufficient for any mutating operation: an adversary who knows m can name the Identity but cannot impersonate it. This separation is only as good as the registration verifier’s refusal to accept a proof from a party holding m but not σ . The A’ registration proof extracts a plain PS signature on m from any accepting transcript (*Proofs*, Theorem R-sound), so registering without the credential is a PS forgery; the earlier verifier accepted a copied public credential with a fresh proof from anyone who knew m , which would have let a prior counterparty register, deposit and mint in another person’s name. A *payment* moves value v from payer P to recipient D .

6.1 Adversaries and privacy goals

The model’s three properties: no one can reconstruct who paid whom from the chain; nothing on chain reveals either party’s Identity to outsiders; and the two parties can always identify each other. The first two must hold against the strongest adversary available: the issuer itself, which knows every identity it ever certified.

Definition 1. *Three properties, held against adversaries of increasing power:*

- (**Bulk-tracing resistance.**) *Mallory, a passive observer with full chain history and polynomial-time computation but no party’s secrets, cannot reconstruct who paid whom, how much, or link a mint to its spend(s).*
- (**Third-party non-disclosure.**) *Nothing on chain reveals either party’s Identity point M to a non-participant, even in aggregate across many payments (including to the issuer, below).*

- (**Mutual decryptability / accountability.**) Each of the two participants, from data they hold plus public chain state, can produce a sound receipt naming the other’s registered M ; and a colluding pair cannot arrange an accepted payment for which no such receipt exists.

The first two properties must hold against the strongest realistic deanonymiser, the **Issuer**: the KYC trust anchor that holds, for every client it certified, the plaintext identity (id_i, m_i, M_i) and the credential σ_i it issued, strictly more than Mallory knows. We treat a compelled or wholly breached issuer in Sec. 6.3.

6.2 The non-deniable-receipt invariant

One definition carries the accountability half of the paper. Its first two clauses are standard: receipts can always be produced, and never forged. The third is the new one: even a payer and payee working together cannot make an accepted payment that leaves no receipt.

Definition 2. (Non-deniable receipt.) *A protocol satisfies the **non-deniable-receipt invariant** if for every accepted payment there exist a receipt $\mathcal{R}$ and an efficient public predicate RcptVerify such that:*

- (**Completeness.**) *The recipient D can compute $\mathcal{R}$ from data they hold plus public chain state, and $\text{RcptVerify}(\mathcal{R}) = 1$ names a registered Identity M_P and value v .*
- (**Soundness / non-frameability.**) *No PPT adversary produces $\mathcal{R}'$ with $\text{RcptVerify}(\mathcal{R}') = 1$ naming an Identity that did not author the payment.*
- (**No deniability.**) *No PPT payer can cause an accepted payment to D for which no accepting receipt naming the payer exists – even in collusion with D .*

Completeness alone protects a self-interested recipient; the third clause is what forbids a *willing* pair. Symmetric statements bind the recipient’s identity toward the payer (Theorem 12 gives the per-flavour content).

6.3 Issuer-blindness: accountability without a bulk opener

The KYC issuer is the system’s only trust anchor, and the natural objection to any *accountable privacy* claim is that the issuer is a bulk-deanonymisation backdoor. It is not. Issuance is *address-blind*: the issuer signs the identity scalar $m = H(id)$ once (a PS credential over m alone) and is thereafter done. The client derives each on-chain account *offline* (a fresh key pair (sk, pk) , a self-formed ElGamal credential $E_{\text{addr}} = (rG, M + rpk)$, and a blinded *presentation* (A, B) of the credential with its proof π), without the issuer ever seeing pk . The issuer’s view is therefore, per certified client i , the tuple $(id_i, m_i, M_i, \sigma_i)$ (the plaintext identity) plus public chain state. It holds no chain addresses.

Theorem 3. (Issuer-blindness.) *Under the hiding of the A ’ registration presentation (Proofs, Theorem 2’: for fresh uniform (a, b) the published (A, B) and the point P its proof reveals are independent of m), zero knowledge of the registration NIZK in the random-oracle model, and IND-CPA security of ElGamal on $\mathbb{G}_1$ (equivalently DDH), an issuer (even one compelled or wholly breached, in possession of $\{(id_i, m_i, M_i, \sigma_i)\}_i$ and the entire chain) has negligible advantage in matching any issued identity M_i to its on-chain registration $(pk_j, E_j, A_j, B_j, \pi_j)$. Compulsion or breach yields the set of certified identities but no map to chain accounts; deanonymisation remains per-account, requiring that account’s sk or a counterparty receipt.*

Proof. Two locks. *Credential lock.* The transcript (A_j, B_j, π_j) is simulatable in the random-oracle model from $(A_j, B_j, P_j, pk_j, E_j)$ alone, and (A_j, B_j, P_j) is distributed as $(U_1, xU_1 + yU_2, U_2)$ whatever m is (*Proofs*, Theorems 2' and R-HVZK); so it carries no information about which certified m_i it presents, even to a party holding (x, y) and every m_i . (The earlier form of this lock, "statistical unlinkability of PS rerandomisation", was wrong: a rerandomised signature is a valid signature on m , which an issuer holding every m_i verifies directly.) *Plaintext lock.* The issuer *knows* M_i , so its task is not decryption but *plaintext-checking*: does $E_j = (R, C)$ encrypt M_i ? That is deciding whether $(G, pk_j, R, C - M_i)$ is a DDH tuple, and an algorithm answering it for a chosen M_i breaks IND-CPA (submit $m_0 = M_i$, m_1 random). Knowing the plaintext thus confers no advantage. With both locks shut and no stored pk , the issuer is reduced to an independent DDH decision per (identity, account) pair. $\square$

This is what *no trusted opener* means here, with the A' presentation in place: the closest thing to an authority cannot open transactions in bulk, even against itself. Under compulsion it can only *verify* a link some party surrenders – map a receipt-disclosed M to the real person – never *produce* one. Two caveats bound the theorem:

- **Computational, not information-theoretic.** The plaintext lock is DDH; only the credential lock is information-theoretic (and its argument is pending independent review). A discrete-log break – in particular a quantum computer running Shor against $\mathbb{G}_1$ – turns the harvested $\{M_i\}$ plus the archived chain into a wholesale, retroactive deanonymisation of *past* identity-bound traffic. The breached or compelled issuer's KYC records are precisely that harvest corpus. Migration of the identity substrate ahead of cryptanalytic risk is a deployment obligation, not an afterthought (Sec. 13).
- **Retroactive, not prospective.** Address-blind issuance protects *past* registrations, under DDH, against *future* issuer compromise; but a coerced issuer can change the protocol going forward (demand pk at KYC, record addresses, backdoor credentials). Issuer-blindness is a statement about issuances already performed under the honest protocol.

6.4 The compulsion calculus

The invariant and issuer-blindness, composed, induce a precise answer to the question *who can a court reach, and what happens when they refuse?* We state it as three properties. Call a party's *opening material* for a payment the secrets from which an accepting receipt can be computed: the note opening (its randomness ρ and identity payload) together with, per flavour, the holder's identity scalar or account key needed to decrypt the counterparty ciphertext.

Proposition 4. (Existence and exclusivity.) *For every accepted payment: (i) accepting receipts naming each counterparty exist and are computable by the other counterparty from material they hold (Def. 2, clauses 1 and 3; Theorem 12); and (ii) no PPT party outside the payment's direct participants – including a coalition of Mallory and the compelled issuer of Theorem 3 – can compute any accepting receipt for it, under the privacy properties of Theorem 15.*

Proposition 5. (No institutional bypass.) *The only compellable institution, the KYC issuer, can perform exactly one useful act under compulsion: given an identity point M already disclosed by a participant's receipt, it maps M to the certified legal person. It cannot – even fully breached – enumerate, sample, or statistically approximate the payments of any person (Theorem 3; computational, and conditional on the A' presentation's review). No other system component (contracts, verifiers, provers, relayers) holds any secret at all.*

Proposition 6. (Refusal-resilience.) *If every direct participant in a payment withholds or destroys their opening material, then no PPT coalition of all remaining actors – the compelled issuer with its full KYC corpus, all chain history, all registry state – gains non-negligible advantage in identifying either party or linking the payment’s mint to its spend. Destruction is irreversible: the opening material, the account secrets sk and the presentation bindings b are the only trapdoors, and none ever existed anywhere but in the participants’ custody.*

Propositions 4–6 follow from the privacy theorems (Sec. 10) plus issuer-blindness. Their content is the *composition* – constitutional design more than cryptography:

- **Due process is fully served.** Evidence exists for every payment (no colluding pair can prevent it), it convicts only the guilty (non-frameability: a receipt cannot be forged against an innocent party; Corollary 14), and a court always has a *specific person* to direct an order at. Investigation proceeds the way warrants are supposed to work: from a known participant, one named counterparty at a time, each step producing sound evidence.
- **Bulk surveillance is not for sale.** There is no database to subpoena, breach, or buy. The marginal cost of surveilling one more citizen is coercing one more citizen: publicly, individually, through legal process directed at a person who can see it happening. The system makes population-scale financial monitoring not illegal but *unimplementable* – a claim that rests on Theorem 3, hence on the A’ presentation and on DDH.
- **The last word belongs to the citizen.** Refusal has personal legal cost (contempt, adverse inference), as it should, under legitimate process. But it is *effective*: a citizen facing an unjust order, or a population facing a judiciary that has begun criminalising beliefs, can withhold or destroy keys, and the archived ledger stays dark regardless of what any institution is later compelled to do. The privacy floor survives institutional capture because no institution was ever load-bearing.

Two honest edges. First, receipts are bilateral: your counterparty’s copy is beyond your control, *by design*. A criminal cannot silence their victim’s receipt by destroying their own. Refusal protects a payment only when every participant elects it; exactly the case the property is for: consensual acts the regime of the day has criminalised. Second, destroying your keys also destroys *your* evidence: the standing ability to prove what you paid and to whom. Refusal-resilience is an option with a price, chosen by its holder – not a default that degrades accountability for anyone else (Def. 2 holds for every accepted payment regardless).

7 The Identity-M Principle

The properties above are only as good as the binding that produces the receipt, and the central design decision is *what* a payment binds and *at which moment*.

The obvious answer—bind to the counterparty’s account public key, the thing that actually signs transactions—fails for notes. (See 3.) An early design bound addressed notes to the recipient’s mint-time key pair; losing that key stranded the note forever, and key rotation broke addressing. More fundamentally, when the two anchors are split across time (payer named only at mint, payee only at spend) and unlinkability forbids any public bridge, binding to a disposable key is not merely brittle but *vacuous*: an ElGamal ciphertext does not commit to its key, so "encrypted under the same key" constraints can be satisfied with a bogus key while the intended recipient can still spend (see Lemma 16). Account keys also cannot survive the gap between an anonymous payer today and an anonymous payee tomorrow.

The Fountain (Sec. 4) gives us a different kind of anchor: the stable, scarce, non-grindable identity point M .

The correct authority axis is therefore the **identity point** $M = H(\text{id})G$:

- **Identities are scarce and non-grindable; keys are free.** Anyone can mint key pairs at will, so key possession proves nothing about *who*. A registered M exists only by KYC certification, and its preimage structure denies an adversary any choice of its value:

Definition 7. (Non-grindable identities.) *Each registered identity point is $M = H(\text{id}) \cdot G$, where id is the KYC-verified attribute record and H (keccak256 over a canonical encoding) is preimage-resistant, modelled as a random oracle. No PPT adversary registers an identity point of its choosing; in particular it cannot register an account whose M equals a prescribed target $M' + \delta G$ for an adversarially chosen $\delta \neq 0$.*

The assumption forbids registering a point of the adversary’s choosing; it does not by itself forbid registering under an *existing* M whose scalar the adversary has learned, and every counterparty learns m . That is prevented by the credential: registration requires the A' presentation with its proof, which extracts a PS signature on m (*Proofs*, Theorem R-sound), and identity disclosure confers no such signature (Sec. 7).

- **Accounts are envelopes; the identity is the addressee.** A note addressed to M_{rec} is spendable from *any* account whose registration binds it to m_{rec} – including one created after the note was minted. Key loss is recovered by registering a fresh account to the same identity; no custodian, no social recovery, no issuer involvement. Dually, the receipt names the *person* (their certified M), not a disposable key.
- **Identity binding survives the time gap.** The deep obstruction in doubly-anonymous payments (Sec. 11) is that the payer’s anchor exists only at mint and the recipient’s only at spend; no single on-chain moment sees both. Account keys cannot bridge the gap (the recipient’s spending key may not exist yet at mint). The identity point can: it is stable across the gap, and – the technical heart of the system – *everything a note must say about its parties can be said as an ElGamal encryption statement whose plaintexts are identity points*.

If the goal is to name a specific registered person to that person (and only that person) without naming them to the pool or the public, the natural cryptographic move is to hand them a ciphertext that only they can open. When the name must be certified once by KYC and the proof must survive

unlinkability across a time gap, the move becomes a re-encryption of the certified point under the recipient's (or issuer's) public key. That is exactly the primitive the system uses.

The last observation collapses the construction into **one gadget**:

The gadget. Prove, in zero knowledge, that a ciphertext is an encryption (or re-encryption) of a *registered* Identity under a target key – where "registered" is certified by membership in the registry accumulator rt_{id} , proven over the same witness as the facts it qualifies, never inferred across two proofs that merely share a public point.

Three instantiations of the gadget – differing only in *which* identity is encrypted under *whose* point, and at which moment – produce the three realisable flavours (Table in Sec. 8). The same inversion gives the spend path its shape: at deposit, the depositor proves *their account is bound to the identity the note addresses*. Reading the note and owning the account rest on two secrets, k and m , so the statement is one SNARK (a succinct zero-knowledge proof, constant cost to verify) over one witness: the account's registered ciphertext decrypts to $M = mG$, a certified leaf pairs m with the key that reads the note, and the whole gate is about *this* note. Authorization keys on the identity; the account merely carries it. And because the identity scalar m conveys no write authority – only the PS credential σ , presented under the A' proof, can authorize account registration or spend – m is safe to disclose to every counterparty: the receipt names a person, not a capability.

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8 Construction

The construction in one paragraph: a note is a hidden commitment in a shared pool. Minting proves the pool received the money and that the note names its issuer. Spending proves – without revealing which note – that some note is yours, that your account is bound to the identity it addresses, that this identity is registered, and that all of these statements are about the *same* note.

The starting point is the system’s ordinary transfer: a synchronous handshake in which each party re-encrypts their credential under the other’s key and proves (Chaum-Pedersen) the re-encryption carries the same M . Both identity anchors are co-present in one call; mutual decryptability is free. A Note reorganises the same handshake through a commitment pool, splitting it across time so that mint and spend are unlinkable – and the rest of the construction is the work of carrying the bilateral binding across that split.

8.1 The pool, the commitment, the nullifier

The pool is a Tornado-style¹⁴ Merkle tree of note commitments, hashed with Poseidon (a hash designed to be cheap inside proof circuits), keeping a rolling window of recent roots and a set of spent *nullifiers*¹⁵—one-way serial numbers, revealed only at spend, that make a double-spend detectable without linking the spend to its note. with Poseidon (a hash designed to be cheap inside proof circuits), keeping a rolling window of recent roots and a set of spent *nullifiers* – one-way serial numbers, revealed only at spend, that make a double-spend detectable without linking the spend to its note. A note is the opening tuple; its on-chain commitment is

$$cm = \text{Poseidon}_6(T_{CM}, \text{flavor}, v, \rho, idHash, \text{predicate}),$$

where ρ is the note’s secret randomness and *idHash* is a Poseidon hash of the flavour’s *identity payload* – the commitment to everything the note says about its parties. Each note hash leads with its own domain tag (T_{CM}, T_{ID}, T_{NF}), a fixed field word that keeps it disjoint from every other hash in the protocol. Batch mints are proven by a Groth16 SNARK (proofs a few hundred bytes, constant gas to verify; one circuit per pinned batch size N) that opens every commitment, range-checks the values, sums them to the public escrowed total, and folds the leaves into the attested new root. The contract escrows exactly the proven total in the same atomic call.

Spends are proven by a single shared, flavour-bound circuit: the prover shows knowledge of an opening of *some* tree leaf under a recent root, with public (*root*, *nf*, *face*, *recipient*, *chainid*, *flavor*, *issuanceCommitment*). The entrypoint supplies its flavor, the circuit requires it to match the committed opening, and the current predicate is constrained to zero. For B1 only, *issuanceCommitment* = cm ; A1/A2 expose zero. The nullifier

$$nf = \text{Poseidon}_3(T_{NF}, \rho, idHash)$$

is one derivation for every flavour (the tag keeps nullifier preimages distinct from commitment preimages). Double-spends are rejected by set membership. No account key enters the hash: spend *authorization* is layered per flavour on top of the nullifier. That is what makes key-loss recovery and account-agnostic deposit possible (Sec. 7) – and what creates the binding obligation the rest of this section meets.

¹⁴A privacy mixer using Merkle trees of commitments and nullifiers to hide which input corresponds to which output, as pioneered by Tornado Cash.

¹⁵A one-way serial number derived from the note’s secret randomness; it lets the system detect double-spends while preserving unlinkability.

8.2 Three flavours, one gadget

What *idHash* commits, and which direction of the gadget runs at which moment, defines the flavour:

Flavour	idHash commits	Issuer named...	Recipient named...
A1	$(eNote, m_{iss})$	at mint, in the clear (batch Schnorr)	at spend: the folded gate, to M_{rec}
A2	$(eNote, eIss, T)$	at spend, to recipient alone: decrypt $eIss$	at spend: the same gate, plus the issuer's membership
B1	(m_{iss})	at mint, in the clear	at spend: depositor binding $eDepForIss$ to issuer

with the per-flavour ciphertexts, all ElGamal. Addressed notes are encrypted to the recipient's receiving key $pk_{recv} = kG$, derived from wallet seed material and independent of the Identity: the identity scalar is disclosed to every counterparty by design, so nothing may be encrypted under M . The registry certifies the pair in one salted accumulator leaf, Poseidon($\tau_R, m, k, salt$).

- **A2 (addressed, private issuer).** The issuer encrypts *its own* registered identity under the recipient's receiving key: $eIss = (r'G, M_I + r'pk_{recv})$, decryptable by k alone; the note value rides in $eNote = (r_nG, vG + r_npk_{recv})$. At mint, a blinded re-encryption sigma (**verifyIssuerReenc**) proves $eIss$ carries the *minting account's registered* identity – the anti-framing anchor – without revealing it, and publishes $T = r'pk_{recv} + \gamma H$ on a generator H hashed to the curve. Each leaf's $eIss$ and T are public inputs of the A2 mint SNARK (the *leaf-tie*), which recomputes *idHash* over them: no leaf can lack a binding, and no binding can float to another leaf. γ reaches the recipient in the delivery.
- **A1 (addressed, public issuer).** The issuer is named openly (a Schnorr signature over the batch under their registered key, checked at mint), and the recipient's identity is encrypted under the same key: $eRec = (r'G, M_{rec} + r'pk_{recv})$.
- **B1 (bearer, public issuer).** No recipient exists at mint (Theorem 8 makes this the *only* coherent bearer cell); at spend the depositor encrypts their own identity under the public issuer's key, $eDepForIss = (rG, M_{dep} + rpk_{iss})$, and proves it carries the same identity their payout account is bound to. The public-mint batch Schnorr authenticates the issuer and **Notes** records every exact commitment under that issuer. At spend the circuit-revealed B1 commitment must map to the issuer used by this depositor-binding proof, so an otherwise-valid substitute issuer cannot compose with the note.

8.3 The spend-side composition

Every addressed deposit (**spendCoupledA1**, **spendCoupledA2**) verifies, in one transaction, the shared flavour-bound note proof plus ONE folded gate (**deposit_fold_a1**, **deposit_fold_a2**, Groth16) carrying every relation over a single witness:

1. The receiving secret k decrypts the note's ciphertext to a point, and the supplied $eEnc$ re-encrypts that same point under the same k .
2. The depositing account **msg.sender** is registered, and its credential decrypts under its own key to the Identity $M = mG$.
3. A registered accumulator leaf commits the PAIR (m, k) under the holder's salt.
4. That leaf's path folds to the posted root rt_{id} .
5. For A2 only: the decrypted ISSUER Identity is itself registered, under the salt the note shipped.

6. For A2 only: the committed T opens as $(r'k)G + \gamma H$ – the key the mint binding proved about is the key the recipient holds (the *key tie*).

The posted root is one the registry retains in its ring and accepts for the Notes consumer (posted within seven days); a membership path is 32 levels, a registry’s subtree and then the aggregator. The composition is one proof and not three because the two facts an addressed spend must establish rest on two DIFFERENT secrets – the receiving key reads the note, the Identity scalar owns the account. Three gates sharing a public point would let a thief with a stolen payload answer the reading half with the stolen key and the Identity half with its own registered Identity: both halves true, nothing joining them. An equality inferred from two proofs that merely share a point is not an equality, and relation (3) is what states it.

```
// inside Notes._spendCoupled (abridged; A2 passes allLayout=false, A1 true):
require(spendVerifier.verifySpend(proof, root, nullifier, face, recipient, chainid));
require(identityRegistry.acceptsRoot(identityRoot, NOTES_MEMBERSHIP_CONSUMER));
require(allLayout
  ? depositFoldVerifier.verifyFoldA1(foldProof, nullifier, face, identityRoot, eEnc, msg.sender)
  : depositFoldVerifier.verifyFoldA2(foldProof, nullifier, identityRoot, eEnc, msg.sender),
  "bad folded deposit gate");
```

The composition discipline – every public input derived on chain from the same objects (`msg.sender`’s registered key and credential read from the registry, $eEnc$, the nullifier, the face, the posted root), never taken from the prover’s bytes – is what makes the relations impossible to decouple. We commend the pattern to designers beyond this system, together with its sharper form: where two facts rest on two secrets, do not compose proofs – fold them. For B1 a depositor-binding sigma (an Okamoto-style proof coupling the payout account, $eDepForIss$, and $P_{dep} = M_{dep} + bH$) composes with a bound membership proof instead, and that IS sound: B1’s two facts rest on one secret, so the sigma’s shared challenge is a genuine tie, and the blind sits on a generator whose logarithm is unknown.

For A2, $eEnc$ is a fresh *re-encryption* of the committed $eIss$ (same plaintext, new randomness): re-using the mint-time ciphertext – a public input of `Notes.mint` – would link spend to mint and destroy the flavour’s anonymity. Hence the tie proves re-encryption *equivalence*, never equality. For A1, the committed payload has no second ciphertext, so the tie runs through the note’s own value ciphertext with the face public; the next subsection explains why that public face is not incidental but the soundness linchpin.

8.4 The A1 tie and the face-pinning principle

ElGamal ciphertexts are not key-committing: given the randomness r_n (which travels with the opening), the key of $eNote = (r_n G, C_n)$ can be re-explained as any k' by absorbing the difference into a free plaintext, $V' = C_n - (r_n k')G$. A binding circuit that witnesses the plaintext freely therefore proves nothing about the addressed mailbox. The A1 tie pins the plaintext *publicly*: its statement fixes $eNote.C = (v + r_n k)G$ with v a public input that the contract sets to the spend’s **face** – itself bound to the note’s committed value by the spend SNARK over the same nullifier. With v and $r_n = \log_G eNote.R$ determined,

$$k = (\log_G eNote.C - v) \cdot r_n^{-1}$$

is *unique*: only the mailbox the note was addressed to satisfies the relation, and relation (3) then says whose Identity that mailbox is (Theorem 10, A1 clause; executable witness in Appendix 14). The observation generalises: when a binding must travel through a non-committing encryption, *publishing the plaintext’s already-public component is the cheapest commitment*.

8.5 The receipt, assembled

A receipt is a four-link proof chain, every link checkable by a third party from public state plus the holder's disclosed material: (a) *opening* – the disclosed tuple re-hashes to a commitment; (b) *minted* – that commitment is a leaf under a historical root; (c) *issuer* – the flavour's issuer binding (Schnorr in the clear, or the decrypted-and-membership-certified *eIss*); (d) *paid* – the nullifier consumed and face transferred. The recipient-naming direction is the spend-side composition above. Link (c) is where the invariant lives or dies; Sec. 11 is about keeping it alive when the issuer is anonymous.

Because the material behind the chain is held by *both* parties – the issuer created the identity payload at mint, the recipient was handed it with the note, and the spend event publishes the rest – either party can assemble the receipt: the depositor's copy and the issuer's "I paid X" copy differ only in who proves their own account-to-identity tie. The addressed legs are proven by whoever holds the evidence – the recipient by verifiable decryption under its receiving key, the issuer by disclosing the mint randomness, which re-encrypts to the committed ciphertexts – and never by decrypting under a scalar a verifier could derive from a disclosed record, which would let anyone who ever saw a receipt read the payee's mail. Both copies check the same committed payload, which is what makes them verify identically. The end-user serialization of this chain – a printable, re-scannable slip – is Sec. 12.2.

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9 The Forced Taxonomy

Who can be named, and how, is fixed by what is known at mint time – the axis that defines the flavours. One of the four nominal cells (bearer $\times$ private issuer, "B2") is not merely undesirable but impossible.

Theorem 8. (No private-issuer bearer note.) *No bearer note can both (i) deliver its issuer's identity to its eventual redeemer (mutual decryptability) and (ii) withhold its issuer's identity from non-redeeming holders and observers (issuer privacy).*

Proof. The argument is information-theoretic. A bearer note's redeemer is unknown at mint: it is whoever comes to hold the opening, which circulates among holders by construction. For the redeemer to name the issuer, the note must carry an issuer-identity binding that the redeemer can open; the issuer creates that binding at mint, when the redeemer is unknown, so it must be openable under key material the eventual redeemer will hold. Exactly two sources of such material exist. (1) *The redeemer's own registered secret:* but the issuer cannot target a key or identity point it has never seen, and "whichever member of the public eventually holds the paper" names no point – targeting one would require predicting the redeemer, i.e. the note would be addressed, not bearer. (2) *Note-bound material derived from the opening itself* (e.g. a key derived from ρ): then *every* holder along the chain of custody can open the binding, so the issuer's identity is disclosed to all holders – and a bearer instrument's holders are, in the limit, the public – violating (ii). A binding openable by neither is openable by the redeemer in particular, violating (i). Hence a hidden issuer requires a designated recipient: bearer and private-issuer are mutually exclusive. $\square$

The contrast with A2 is exact: A2 reveals the issuer to *one chosen party* by encrypting under a point only that party's identity scalar opens – which is possible precisely because the recipient is designated. The realisable set is therefore $\{A1, A2, B1\}$, and the implementation enforces the boundary structurally: the mint circuit's per-leaf issuer-mode signal admits no private-issuer bearer codepoint, so a B2 leaf is not merely rejected but *unprovable*.

10 Security

Six results carry the system: the deposit gate is sound (a depositor really is the identity the note's mailbox belongs to); the tie is sound (the claim is about the very note being spent); the A2 key tie is sound (the issuer a recipient decrypts is the minter, not a registered puppet); every payment yields receipts in both directions; no coalition can avoid them; and nothing else leaks. The only available cheat is self-sabotage: a coalition can un-name its own note – making it unspendable – never frame an innocent. Full reductions are Theorems 1–12 of the companion *Proofs* document; each result here carries an executable witness (Appendix 14). Assumptions: discrete log / DDH on $\text{BN254 } \mathbb{G}_1$; Poseidon collision-resistance and PRF behaviour (ROM); Groth16 knowledge-soundness and zero-knowledge; Schnorr/Okamoto special soundness; a generator H hashed to the curve, so that no one knows $\log_G H$; and Assumption 7.

Theorem 9. (Deposit gate soundness.) *If an addressed spend is accepted, then except with negligible probability there exist extractable $(m, k, sk_{\text{dep}}, \text{salt})$ and a 32-level path such that: the depositing account is registered, under the A' proof, with a credential decrypting to $M = mG$; a certified leaf $\text{Poseidon}(\tau_R, m, k, \text{salt})$ folds to an accepted posted root rt_{id} ; and k decrypts the supplied $e\text{Enc}$ to the note's point. One m and one k signal enter every relation, so a thief holding a stolen payload – and so k – has no witness under its own identity. For a bearer spend, the depositor-binding sigma and the membership proof open one $P_{\text{dep}} = M + bH$, and answering them with two identities requires $\log_G H$.*

Theorem 10. (Note $\leftrightarrow$ eEnc tie.) *If the binding proof is accepted for nullifier nf , then except with negligible probability the extractor produces an opening $(\rho, \text{idHash}, \dots)$ with $nf = \text{Poseidon}_3(T_{NF}, \rho, \text{idHash})$ – the very nullifier the spend consumed – and:*

- **(A2 layout.)** $\text{idHash} = \text{Poseidon}_{11}(T_{ID}, e\text{Note}, e\text{Iss}_0, T)$, the committed $e\text{Iss}_0$ decrypts under k to M_I , the supplied $e\text{Enc}$ is a re-randomisation of it under the same k , and M_I 's salted leaf folds to the same root.
- **(A1 layout.)** $\text{idHash} = \text{Poseidon}_6(T_{ID}, e\text{Note}, m_{\text{iss}})$, $e\text{Note} = (r_n G, (v + r_n k)G)$ with v the spend's public face, and $e\text{Enc}$ is a fresh encryption of M_{rec} under k . The public v makes k unique (Sec. 8); a holder of a stolen opening with its own certified pair (m', k') has no satisfying witness.

These are relations of the same folded proof as Theorem 9, so the depositor's identity is the identity the spent note's mailbox belongs to, and (A2) the membership-certified point is the committed ciphertext's decryption.

Theorem 11. (The A2 key tie.) *If an A2 note was minted under an accepted re-encryption binding and spent under an accepted fold, then except with negligible probability the issuer identity the fold certifies is the minting account's registered identity. The binding extracts $T = r' pk_Q + \gamma H$ and $C_i = M_{\text{iss}} + r' pk_Q$ for a key pk_Q the minter chose; the fold extracts $T = r' kG + \gamma' H$ and $C_i = M_I + r' kG$ over the same committed $(e\text{Iss}, T)$; so $M_{\text{iss}} - M_I = (\gamma - \gamma')H$, and $M_I \neq M_{\text{iss}}$ for two registered identities yields $\log_G H$. A receipt checks the same fact as $C_i - T + \gamma H = M_{\text{named}}$.*

Theorem 12. (Mutual decryptability, all flavours.) *For every accepted payment, each party can produce an accepting receipt naming the other's registered identity: A1/B1 recipients name the issuer from the in-clear M_{iss} plus batch Schnorr; A2 recipients decrypt $e\text{Iss}$ under k , with the mint binding, the key tie and membership certifying it is the registered minter; A1/A2 issuers hold the addressed M_{rec} from mint and a mailbox leaf tying pk_{recv} to it; B1 issuers decrypt $e\text{DepForIss}$. All receipts are third-party checkable. A receipt names a person only given unforgeable registration (Sec. 7).*

Theorem 13. (No deniability under collusion.) *No PPT payer/recipient coalition produces an accepted payment with no accepting receipt. At mint, every leaf must carry a verified issuer binding (Schnorr or blinded re-encryption + leaf-tie; the circuit’s issuer-mode signal leaves no unbound codepoint). At spend, the deposit gate (Thms. 9, 10) reverts unless the verified ciphertext – provably the note’s own – decrypts, under the key certified beside the depositor’s identity, to a registry member. A coalition can mint a note whose committed identity material is garbage, but such a note admits no membership witness and is unspendable: un-nameable implies un-spendable.*

Corollary 14. (Deny, never frame.) *The only deviation available to a coalition is to render a note un-nameable – in which case it cannot be spent – never to make a receipt name an innocent third party: every binding is proven over msg.sender ’s own registered credential (mint) or the depositor’s own account binding (spend); registration itself is unforgeable without the credential (the A' proof extracts a PS signature on m , so a prior counterparty who learned an innocent party’s m cannot register in their name – without this premise the corollary would be vacuous); the key tie (Theorem 11) denies a minter a registered puppet in its own place; and Assumption 7 denies the coalition any registered identity at an offset of its choosing. Receipts are therefore evidence-grade: an accepting receipt names a party that actually acted.*

Theorem 15. (Privacy.) *Against Mallory and against the (even breached) issuer: commitments and nullifiers reveal nothing (Poseidon hiding/PRF); a spend links to its mint only through n_f , whose preimage includes the secret p ; the folded gates are Groth16 zero-knowledge with public inputs derived from already-public values; $B1$ ’s sigma is HVZK and its only identity-derived public value is the perfectly-hiding $P_{\text{dep}} = M + bH$; $e\text{Enc}$ is a uniform re-randomisation under an unpublished key; and the accumulator leaves are salted, so a party holding every certified identity scalar decides no membership. The residual public surface of a spend is exactly (face, recipient account) – mitigated operationally by fixed denominations and relayed payouts – and of a mint, the batch total.*

Proof sketches: Theorem 9 is Groth16 knowledge-soundness over one witness, its relations joined by signal identity rather than by inference across proofs; its bearer clause is Okamoto special soundness plus Groth16 knowledge-soundness over one P_{dep} (any discrepancy yields $\log_G H$). Theorem 10 is the same extraction, with Poseidon collision resistance joining the fold’s idHash to the spend’s nullifier; the $A1$ uniqueness clause is the face-pinning argument. Theorem 11 combines the binding’s and the fold’s extractors over the committed T . Theorems 12–15 and the corollary compose these with the mint bindings and the identity-layer results (the hiding of the registration presentation, *Proofs* Theorem 2’; ElGamal IND-CPA; Theorem 3). Full statements: *Proofs* Theorems 7–12.

11 The Cost of Anonymity

The public-issuer flavours bind the issuer with a signature and the depositor with sigma protocols (cheap EIP-196 arithmetic, no SNARK over identities beyond the shared membership proof). What makes the doubly-anonymous flavour (A2: issuer named only at mint, recipient only at spend, the two unlinkable) categorically harder? In short: when nobody is named in the clear at the binding moment, the cheap bindings prove nothing, and a zero-knowledge proof of registry membership becomes unavoidable. This section locates that boundary: the diagnosis, the vacuity of the obvious fix, the necessity result, and the shipped architecture as the branch the necessity result permits.

11.1 The EOA-transfer mirror: the cost *is* the anonymity

A direct transfer between registered accounts gets collusion-resistance free: the mandatory approve handshake reads *both* parties' registered records in a single call – both anchors co-present, both named. A2 splits the anchors in time *by demand of its privacy goal*: the issuer's anchor is live only at mint, the recipient's only at spend, and unlinkability forbids any public value from bridging them. The A2 mint binding is, line for line, the approve handshake with one change: the recipient's identity is *blinded*. That one change is the entire gap – it surrenders the phrase "under the recipient's *registered* key" – and everything that follows is the price of buying the phrase back without unblinding anything.

11.2 The obvious fix is vacuous

Lemma 16. (Vacuity of key-equality coupling.) *Let an A2 mint constrain only that the issuer payload $eIss$ and the note ciphertext E_{note} are encrypted under the same key pk_{rec} , with the recipient's identity point M_{rec} a free witness. Then for any issuer-chosen pk_{bogus} the constraint is satisfiable while the note remains spendable by the intended recipient D and $eIss$ is encrypted under $pk_{bogus} \neq pk_D$.*

Proof. Set $pk_{rec} := pk_{bogus}$ and $M_{rec} := M_D + r_n(pk_D - pk_{bogus})$. Then $E_{note} = (r_nG, M_{rec} + r_npk_{rec}) = (r_nG, M_D + r_npk_D)$, a valid encryption of M_D under D 's registered key, so the spend-side check accepts and D spends; while $eIss = (r'G, M_{iss} + r'pk_{bogus})$ is under pk_{bogus} , so a compelled sk_D decrypts $eIss$ to $M_{iss} + r'(pk_{bogus} - pk_D) \neq M_{iss}$. The free witness M_{rec} absorbs the constraint. (Executable witness: Appendix 14.) $\square$

The lemma is the formal autopsy of an entire class of cheap designs, and the second motivation (after key loss) for the Identity-M inversion: binding to a key is worthless, because keys are free and ElGamal does not commit to them. The binding must reach a *registered* identity; the issuer, holding no recipient secret, can assert registration only as a set-membership statement.

11.3 Necessity: set-membership, at one of exactly two moments

Proposition 17. (Conditional necessity.) *Let Π be a protocol that, in the A2 setting, (i) preserves bulk-tracing resistance and third-party non-disclosure (issuer named only at mint, recipient only at spend, mint and spend unlinkable), (ii) closes the no-deniability clause, and (iii) achieves the binding by referencing a registered party at a single protocol step. Then Π performs a zero-knowledge proof of membership in the identity registry – over the recipient set at mint, or the issuer set at spend.*

Proof. By Lemma 16, (ii) forces Π to certify that $eIss$ decrypts, under the addressed recipient's registered identity material, to a registered *issuer* identity – binding to some key alone does not

close the clause. This certificate references a registered party: the recipient (at mint), or the issuer that *eIss* re-encrypts (at spend). By (iii) it is a single step’s reference, either in the clear or in zero knowledge. *In the clear is ruled out*: revealing the referenced identity or key on chain names a party and makes them linkable across mint and spend, violating (i); importing the spend-side ciphertext (or a deterministic commitment to it) into the mint to bind there re-links mint and spend, also violating (i). *Zero-knowledge is set-membership*: a reference certifying the referenced party is registered without revealing which is, by definition, a zero-knowledge proof of membership in the registry. As (i) excludes the in-clear case, Π performs this. $\square$

11.4 The shipped system is the predicted branch

Proposition 17 permits exactly two homes for the membership proof, and the system’s own history visited both. An earlier design (retained in the retired drafting skeleton of this paper) placed it at *mint*: a per-leaf circuit opening the issuer’s credential, proving the recipient’s account-keyed credential a member of an account registry, and coupling the payload under the recipient’s *account key* – at roughly 10^6 R1CS per leaf, on top of the account-pinning brittleness of Sec. 7. The shipped system takes the *spend* branch, and the Identity-M inversion is what makes the branch practical:

- **The membership moves to where its prover is.** At spend the depositor is live, holds m_{rec} and k , and can prove membership of the relevant point – the *issuer’s* decrypted identity for A2 (closing the clause exactly as the proposition demands), the recipient’s own for A1. The proof is per-*spend*, not per-minted-leaf, and proving cost lands on the interested party’s own hardware; the chain only ever verifies.
- **The statement does not split, because the secrets do.** Reading the note and owning the account rest on two different secrets, the receiving key and the identity scalar, and two proofs that share a public point cannot say they belong to one person. So the account relation, the decryption, the certified pair, the membership and the note tie are one folded Groth16 proof (Theorem 9), with every public input derived on chain – the discipline of Sec. 8. Only B1, whose two facts rest on one secret, keeps a sigma beside its membership proof.
- **Relocation creates one new obligation, and it is dischargeable.** Binding at spend means the verified ciphertext is the *depositor’s submission*, so it must be tied to the note being spent – otherwise a stolen opening redeems with a self-addressed ciphertext, and a coalition substitutes a nameable one for an un-nameable commitment. That obligation is Theorem 10, relations of the same folded proof, including the A1 face-pinning trick – and, for A2, the key tie of Theorem 11, since a binding proven at mint must be joined to the key opened at spend through the one channel between them, *idHash*.

The result: the doubly-anonymous flavour, predicted expensive, costs a measured $\sim 70K$ gas more per spend than the bearer flavour (Sec. 12) and seconds of off-chain proving – practical today. Whether restriction (iii) can be dropped – whether a binding amortised across steps, a two-party mint with a blinded recipient contribution, or a witness-encryption construction evades set-membership entirely – remains the paper’s central open problem (Sec. 13): an unconditional impossibility would prove the SNARK necessary; a loophole would be a cheaper design.

11.5 A mechanism remark: the recipient-burden posture

Before the tie relations shipped, the design contemplated an economic backstop: an honest A2 recipient verifies the binding *at delivery* (decrypt *eIss*, check the registered identity) before accepting.

An audit regime with penalty L and probability q then makes verify-then-accept dominant whenever qL exceeds the collusion benefit – and, by Corollary 14, the residual off-equilibrium outcome is only ever an un-nameable note held by an always-identifiable depositor, never a framed third party. With the tie enforced on chain the mechanism no longer carries weight, but we record it: delivery-time verification remains good wallet practice, and the analysis explains why even a deployment that lags the verifier rollout degrades to a bounded, recipient-borne risk rather than an open hole.

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12 Implementation and Evaluation

Everything measured below is shipped, open, and reproducible from the repository: Solidity contracts (**Notes**, **IdentityRegistry**, the generated Groth16 verifiers behind thin adapters), circom circuits, the Python wallet reference, and an end-to-end Foundry suite (**NotesE2E**) that drives one full lifecycle per flavour with *every* verifier the real Groth16 artifact – real batch mint, real spend proof against the replayed tree, and the real deposit gate: one folded proof for the addressed flavours, the depositor-binding sigma and its membership proof for the bearer one.

12.1 What is shipped

Component	Realisation
Identity substrate	IdentityRegistry : ElGamal credentials, blinded PS presentation proofs (A'), a two-level Poseidon accumulator (salted re
Batch mint	mint_batch[a2].circom per pinned N ; issuer Schnorr (public) / blinded re-encryption binding + leaf-tie over $eIss$ and T
Spend	spend.circom (shared, public flavour + $B1$ issuance commitment, note tree depth 20); nullifier Poseidon3(T_NF, rho, id
Deposit gates	deposit_fold_a1 / deposit_fold_a2 (Groth16, addressed, 32-level membership), verifyDepositorBinding + identity_m
Bound membership	identity_membership_b1.circom ($B1$ only); public inputs derived from the sigma's P_{dep}
Domain separation	every transcript, leaf, derived key and the identity scalar carries a AlbertaBuck/.../v2 tag
Note tie	a RELATION of the folded gate, not a proof beside it: the circuit recomputes the nullifier from the $idHash$ it opens
Spend entrypoints	spendCoupledA1 / A2 / B1
Receipts	AB-RCPT/2 envelope + verifier + renderer; all flavours, generated by <i>either</i> Note party (Sec. 12.2)

12.2 The receipt in hand

The invariant’s deliverable is not a theorem but an artifact a citizen can hold. The wallet serializes the verified proof chain as **AB-RCPT/2**: a canonical, bit-reproducible byte string, wrapped into a plain-text payload sized for an 80mm thermal printer – a literal cash-register slip whose face names both parties in the clear and whose payload re-verifies *offline* from its own re-scanned text (`albertabuck verify -`), or anchors to any node for the chain facts.

Two design points carry the civic claims. First, the slip names **persons**, not keys: it embeds each party's full `canonical_identity_data` preimage, and verification recomputes $M = \text{keccak}(\text{data}) \cdot G$ against the named identity point – so "my counterparty is who they say they are" is a hash check, not a promise, and the accounts printed beside the names are pinned by the same proofs (the issuer bindings fold the minting account into their transcripts; the spend event anchors the payout account). Second, for Notes the receipt is **bilateral by construction**: the payload behind the leaf's `idHash` is held by both parties, so the depositor prints their copy and the issuer prints an "I paid X" copy – for B1, naming the depositor by verifiably decrypting the spend event's `eDepForIss` – and both verify identically, because each proves its addressed legs from evidence only it holds, against the same committed payload. Eight golden renders (the five kinds plus the three issuer-side Note receipts) are pinned byte-for-byte in the test suite; the recipient's copy for an A2 note – a payment whose issuer no observer, subpoena, or breached KYC issuer could name – reads:

[illegible]

[illegible]

```
VERIFICATION
  status @ issue: VALID
  receipt  fo6idsp2dmel
```

This is the compulsion calculus (Sec. 6.4) made tangible: standing, court-grade provenance for every payment a citizen was party to – against fraud, theft, or commerce conducted with their stolen credentials – produced from their own records, without anyone’s permission, forgeable by no one; while the same payment remains noise to every non-party, from the chain-analytics firm to the compelled issuer. That pairing is the shift the system exists to make: the capability to *prove* moves to the individuals who transacted, and the capability to *watch* is withdrawn from everyone else. The slip itself is a record, not a bearer instrument: it moves no funds, and discloses only what its holder chooses to disclose, one payment at a time.

Full-call gas for one mint (batch of one) and one coupled spend, per flavour, from the end-to-end suite (chain id 1, Cancun):

Isolated per-phase verification gas:

The cost structure tells the paper’s story in numbers. Both *addressed* flavours pay one folded gate – the price of “only M_{rec} can spend”, within about 70K of the bearer flavour’s sigma and membership – while issuer *anonymity* costs only the mint-side delta ($\sim 116\text{K}$, the blinded re-encryption binding and its T outputs): A2’s spend is within two percent of A1’s. Set-membership, the provably necessary ingredient, costs the same whatever the registry’s size: a 32-level path inside the fold, or one 253K Groth16 verify for B1. Prover side: the folds are 3.31M (A1) and 3.83M (A2) R1CS constraints, dominated by fixed-base scalar multiplications (the A1 fold performs no curve addition at all, since every point it touches is a known multiple of G ; A2’s key tie adds one multiplication on H). They prove in 15–18 seconds under rapidsnark on commodity Apple-silicon hardware, witness generation via the circom C++ calculator. Distribution cost concentrates the same way: the fold *proving keys* are 1.9 GB (A1) and 2.1 GB (A2) one-time wallet downloads, B1’s membership key 272 MB, while every other prover artifact (mint, spend keys and witness generators) is a few megabytes

to a few hundred – and the trusted-setup transcript never ships to wallets at all. All verification is constant-gas on chain; no participant ever waits on another’s prover.

12.4 Verification methodology and artifact

The same vectors flow through three independent implementations – the Python wallet reference (`py_ecc`-style BN254 arithmetic), the circom witness generators, and the Solidity verifiers – and the test suites assert byte-for-byte agreement: over 600 Foundry tests (among them the 12-run all-real-verifier end-to-end lifecycle) and over 450 Python tests (among them byte-pinned golden renders of all eight receipt kinds, from both Note parties’ sides), with the Rust and JavaScript kernels replaying the same vectors. The companion documents are executable: the walkthrough document runs every cryptographic step of all three flavours in one seeded session, and this paper’s appendix witnesses re-run on each build. Honesty items: Groth16 needs a one-time trusted setup whose secret randomness must be destroyed, and ours currently uses development entropy – a production Powers-of-Tau ceremony is a deployment prerequisite. Formal verification of the circuits (beyond witness/constraint testing) is planned, not performed.

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13 Discussion, Limitations, and Future Work

What remains: one open theory question (is the set-membership SNARK truly unavoidable?), a short list of limitations, and the civic argument restated with the system in hand.

Open problem (the necessity lower bound). Proposition 17 is conditional on a single-binding-step restriction. The unconditional question – *must* any A2-private, collusion-resistant protocol perform registry set-membership ZK somewhere? – is open. The escape routes the restriction excludes are precisely the interesting future designs: bindings amortised across multiple steps; two-party mints where the recipient contributes a blinded verifiable-decryption witness without exposing the note ciphertext; witness-encryption-style constructions; and accountability that does not bind *eIss* at all (threshold or selective escrow – a strictly weaker trust model we reject for this system, since it reintroduces an opener).

Limitations.

- The spender’s **face** and recipient account are public (an ERC-20 must pay someone a known amount); mitigations are operational (fixed denominations, relayed payouts), not cryptographic.
- The registration presentation (A’) on which issuer-blindness, non-frameability and the read/write separation now rest is implemented across the four backends but its security argument (*Proofs*, Part I) has not been independently reviewed; until it is, those results are conditional. Its cost is measured: 443,892 gas per registration against 472,630 before it.
- Refusal-resilience and issuer-blindness are computational (DDH); archived ciphertexts are exposed to future cryptanalysis, and the KYC corpus is a harvest-now-decrypt-later target. A quantum-migration path for the identity substrate is future work with a deadline outside our control.
- The KYC issuer is trusted for *certification* (one person, one identity) and for honest-protocol issuance going forward (Sec. 6.3’s prospective caveat); Sybil identities from a corrupt certifier defeat naming, though never framing (Cor. 14).
- Trusted setup: development ptau until a ceremony is run.
- The registry accumulator is append-only in the current deployment; identity revocation/rotation semantics are designed but not shipped.

The autonomy thesis, restated. A jurisdiction adopting this design gets per-instrument compliance *stronger* than physical cash ever offered: every payment carries court-grade evidence, held by exactly the parties a lawful investigation would approach anyway, while forgoing the one thing every CBDC design quietly includes: the standing technical capability of population-scale financial surveillance. We contend the forgone capability is the point. Evidence that must be sought person by person, under process the subject can see and contest and – at personal cost – refuse, is the cryptographic shape of due process. A ledger that stays dark when its citizens withhold their keys is the cryptographic shape of the presumption that free people’s affairs are their own. The mathematics here does not decide which laws are just; it decides only that enforcing them runs through citizens, not over them. And that choice, once deployed, is not revocable by the preferences of any later administration: there is nothing left standing in the middle to compel.

14 Appendix A: Executable Theorem Witnesses

Each witness below executes, in this document’s build, against the same `alberta_buck.wallet` implementation whose vectors drive the circom and Solidity test suites – its curve arithmetic running on the compiled buck-identity kernel (`buck_core.buck_identity`, arkworks BN254; the setup block asserts it, and the pure-Python `py_ecc` path remains the executable specification the kernel is proven bit-identical against). The printed **RESULTS** blocks are regenerated on export (one seeded, deterministic session). They are sanity witnesses, not proofs: each exhibits the object whose existence (or impossibility-of-construction) the named result claims.

14.1 Setup: identities, accounts, registry

```
import random
from alberta_buck.wallet import backend
from alberta_buck.wallet.bn254 import G1, ORDER, mul, add, neg, eq, rand_scalar
from alberta_buck.wallet.elgamal import elgamal_encrypt, elgamal_decrypt
from alberta_buck.wallet.unilateral_a2 import (
    IdentityTree, mint_unilateral_a2, make_receipt, verify_receipt,
)

assert backend() == "kernel", "must run the production kernel: evaluate in the repo venv"
print(f"wallet crypto backend    = {backend()} (buck_core.buck_identity)")

_seed = random.Random(0xACC0)                # one deterministic transcript
rng    = lambda: _seed.getrandbits(256)
CHAINID = 1

def account(m):
    # a registered account bound to identity m
    sk = rand_scalar(rng); pk = mul(G1, sk)
    return dict(m=m, M=mul(G1, m), sk=sk, pk=pk,
                E=elgamal_encrypt(mul(G1, m), pk, rand_scalar(rng)))

def pt(P):
    return f"({int(P[0]) % 10**8:>8d}..., {int(P[1]) % 10**8:>8d}...)"

def dec(R, C, sk):
    # ElGamal decrypt on raw points
    return add(C, neg(mul(R, sk)))

m_iss, m_rec = rand_scalar(rng), rand_scalar(rng)
issuer, rec = account(m_iss), account(m_rec)
M_rec = rec["M"]
k_recv = rand_scalar(rng)                # the MAILBOX secret: independent of m_rec
pk_recv = mul(G1, k_recv)                # what a payer addresses a note to
tree = IdentityTree()
for a in (issuer, rec):
    tree.insert(a["M"])
print(f"issuer Identity    M_I    = {pt(issuer['M'])}")
print(f"recipient Identity M_rec = {pt(M_rec)}")
print(f"recipient mailbox pk_recv = {pt(pk_recv)}")
print(f"registry root      = {hex(tree.root()[20])}...")

wallet_crypto_backend    = kernel (buck_core.buck_identity)
issuer Identity    M_I    = (24518486..., 10011201...)
recipient Identity M_rec = (48294773..., 86086204...)
recipient mailbox pk_recv = (20585238..., 51153208...)
registry root      = 0x274588453a4dd89ec4...
```

14.2 Witness: Lemma 16 (key-equality coupling is vacuous)

The attack object the lemma constructs: a note spendable by D whose issuer payload is keyed to a bogus key, satisfying the naive same-key constraint via the free witness.

```

sk_D, pk_D = rec["sk"], rec["pk"]
M_D        = rec["M"]
pk_bogus   = mul(G1, rand_scalar(rng))
r_n, r_p   = rand_scalar(rng), rand_scalar(rng)

# Free witness absorbs the constraint: M_rec' = M_D + r_n*(pk_D - pk_bogus)
M_rec_free = add(M_D, mul(add(pk_D, neg(pk_bogus)), r_n))
E_note     = (mul(G1, r_n), add(M_rec_free, mul(pk_bogus, r_n))) # "same key" pk_bogus
eIss       = (mul(G1, r_p), add(issuer["M"], mul(pk_bogus, r_p))) # also under pk_bogus

assert eq(dec(*E_note, sk_D), M_D)          # note still decrypts for D: spendable
assert not eq(dec(*eIss, sk_D), issuer["M"]) # compelled sk_D cannot name the issuer
print("naive key-equality coupling: SATISFIED by the adversary, yet")
print(f" E_note decrypts under sk_D to M_D : True (note spendable by D)")
print(f" eIss decrypts under sk_D to M_I : False (receipt destroyed)")
print("-> binding to 'a key' is vacuous; the binding must reach a REGISTERED identity")

naive key-equality coupling: SATISFIED by the adversary, yet
E_note decrypts under sk_D to M_D : True (note spendable by D)
eIss decrypts under sk_D to M_I : False (receipt destroyed)
-> binding to 'a key' is vacuous; the binding must reach a REGISTERED identity

```

14.3 Witness: Theorems 12 / 13 (A2 receipt; un-nameable is un-spendable)

The honest A2 path: the recipient alone produces a receipt naming both parties. The colluding path: an *eIss* keyed to a throwaway point decrypts to garbage that is *not* a registry member – the membership gate has no witness, so the note cannot be spent.

```

note = mint_unilateral_a2(issuer["sk"], issuer["E"], pk_recv,
                          v=1000, rho=rand_scalar(rng),
                          issuer=0xA11CE, chainid=CHAINID, rng=rng)
receipt = make_receipt(k_recv, M_rec, note, issuer=0xA11CE, chainid=CHAINID,
                      tree=tree, rng=rng)
res = verify_receipt(receipt, issuer["pk"], issuer["E"], tree.root(), tree)
assert res.valid and eq(res.issuer_M, issuer["M"]) and eq(res.recipient_M, M_rec)
print(f"A2 receipt: {res.reason} -- names issuer {pt(res.issuer_M)}")
print(f"          and recipient {pt(res.recipient_M)}, value {res.value}")

# Collusion: key eIss to a throwaway point.
pk_throw = mul(G1, rand_scalar(rng))
eIss_bad = elgamal_encrypt(issuer["M"], pk_throw, rand_scalar(rng))
M_garbage = elgamal_decrypt(eIss_bad, k_recv)
print(f"colluding eIss decrypts under k_recv to {pt(M_garbage)}")
print(f" registered identity? {tree.contains(M_garbage)} -> membership unprovable;"
      f" deposit gate reverts: un-nameable => un-spendable")

A2 receipt: VALID -- names issuer (24518486..., 10011201...)
               and recipient (48294773..., 86086204..., value 1000)
colluding eIss decrypts under k_recv to (53767239..., 63884266...)
registered identity? False -> membership unprovable; deposit gate reverts: un-nameable => un-spendable

```

14.4 Witness: Theorem 10 (A2 layout – substitution cannot bind)

The colluders *can* open a bogus ciphertext of their own (they know its randomness) – but the gate does not take the ciphertext as a public input. It opens the *idHash* and recomputes the nullifier from it, and the chain compares that against the nullifier the spend proof consumed. A substituted ciphertext is a different *idHash*, hence a different nullifier, hence a different note.

```

from alberta_buck.wallet.notes import id_hash_a2, nullifier

nf_spent = nullifier(note.opening.rho, note.opening.id_hash)

r_bad = rand_scalar(rng)
eIss_sub = elgamal_encrypt(issuer["M"], pk_recv, r_bad) # nameable substitute
idh_sub = id_hash_a2(note.eNote, eIss_sub, note.binding.T)
nf_sub = nullifier(note.opening.rho, idh_sub)

print(f"spent note's nullifier      = {hex(nf_spent)[:20]}...")
print(f"substituted-note nullifier  = {hex(nf_sub)[:20]}...")
assert nf_sub != nf_spent
print("-> the fold recomputes the nullifier from the idHash it opens, so a")
print("  substituted ciphertext cannot bind THIS spend")

spent note's nullifier      = 0x2df2ed5819b3373096...
substituted-note nullifier  = 0x21d7c61b44d2216b85...
-> the fold recomputes the nullifier from the idHash it opens, so a
    substituted ciphertext cannot bind THIS spend

```

14.5 Witness: Theorem 11 (A2 – a split key names no puppet)

The minter keeps its binding true of its own identity but aims the key so that the recipient's k opens the same $eIss$ to a registered puppet M_B . The committed T then opens under the minter's key, not the recipient's: the fold's relation (6) would need $M_B - M_{iss}$ as a known multiple of H , and the receipt's check names the minter, not the puppet.

```

from alberta_buck.wallet.nums import H_PEDERSEN as H

M_B = mul(G1, rand_scalar(rng)) # the minter's registered puppet
tree.insert(M_B)
r_s, gamma = rand_scalar(rng), rand_scalar(rng)
pk_Q = add(pk_recv, mul(add(M_B, neg(issuer["M"])), pow(r_s, -1, ORDER)))
eIss_s = elgamal_encrypt(issuer["M"], pk_Q, r_s) # the binding speaks of pk_Q
T_s = add(mul(pk_Q, r_s), mul(H, gamma)) # L3, as the minter proves it
print(f"k opens the split eIss to the puppet: {eq(elgamal_decrypt(eIss_s, k_recv), M_B)}")
print(f"and the puppet is registered: {tree.contains(M_B)}")
tie = add(mul(G1, r_s * k_recv % ORDER), mul(H, gamma))
print(f"fold relation (6) T == r*k*G + gamma*H: {eq(T_s, tie)}")
print(f"the gap is exactly M_B - M_iss: {eq(add(T_s, neg(tie)), add(M_B, neg(issuer['M'])))}")
named = add(add(eIss_s.C, neg(T_s)), mul(H, gamma))
print(f"receipt: C_i - T + gamma*H is the minter: {eq(named, issuer['M'])}, "
      f"the puppet: {eq(named, M_B)}")

k opens the split eIss to the puppet: True
and the puppet is registered: True
fold relation (6) T == r*k*G + gamma*H: False
the gap is exactly M_B - M_iss: True
receipt: C_i - T + gamma*H is the minter: True, the puppet: False

```

14.6 Witness: Theorem 10 (A1 layout – the public face pins the identity)

A1's gate pins the note through its VALUE ciphertext with the face public: $eNote = (r_n G, (v + r_n k)G)$. The face leaves no free plaintext to absorb a re-keying, so a thief holding the WHOLE opening – including r_n – still cannot produce the witness, because the relation determines k and the thief does not have it.

```

from alberta_buck.wallet.unilateral_a1 import mint_unilateral_a1

note_a1 = mint_unilateral_a1(M_rec, pk_recv, v=100, rho=rand_scalar(rng),
                             m_issuer=m_iss, rng=rng)

# The relation the circuit enforces, with v PUBLIC.
u_honest = (note_a1.opening.v + note_a1.r_note * k_recv) % ORDER
print(f"addressed: eNote.C == (v + rn*k)*G ->",
      eq(note_a1.eNote.C, mul(G1, u_honest)))

```

```

k_thief = rand_scalar(rng)          # thief holds the whole opening, incl. r_note
u_thief = (note_a1.opening.v + note_a1.r_note * k_thief) % ORDER
print(f"thief:      eNote.C == (v + rn*k')*G ->",
      eq(note_a1.eNote.C, mul(G1, u_thief)))
print("-> the public face pins the plaintext, so only the addressed mailbox key")
print("  closes the relation; relation (3) then pins whose Identity that key is")

```

```

addressed: eNote.C == (v + rn*k)*G -> True
thief:      eNote.C == (v + rn*k')*G -> False
-> the public face pins the plaintext, so only the addressed mailbox key
  closes the relation; relation (3) then pins whose Identity that key is

```

14.7 Reproducing the full system

The on-chain halves of every claim are exercised by the repository's suites; the end-to-end lifecycle (real batch mint, real spend, and the real deposit gate, per flavour) reproduces with:

```

$ make nix-snark-deposit-fold-a1 nix-snark-deposit-fold-a2 # the addressed gates
$ make nix-snark-b1-membership                             # the bearer one
$ make nix-snark-e2e-fixtures                             # all-real-verifier fixtures
$ make nix-match-NotesE2E
[PASS] ... 12 tests: lifecycle, double-spend revert, per-phase gas, x3 flavours

```

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